Jacobian-lens trajectories under medical SFT

`gemma-3-{270m,1b,4b,12b}-it`

August 7, 2026

Full fine-tune on `mix_default`, lr $1\mathrm{e}{-5}$, log-spaced checkpoints to step 4882

Draft — findings 2 and 3 are held pending the 27b arm. All behavioral numbers are on the **current** instrument (`rp`=`1.1`, self-hosted Kimi judge); nothing here is comparable to a version of this deck dated before 2026-08-05.

Setup

- **Four arms**, identical seed and data order: `gemma-3-{270m,1b,4b,12b}-it`, full FT, lr $1\mathrm{e}{-5}$, completion-only loss. A 27b arm is in flight.
- **Log-spaced checkpoints** $\{1, 2, 4, \ldots, 4096\}$ + final (4882). $t = 0$ is the untouched base, prepended by the probe.
- **Lens refit per checkpoint** on a frozen 1000-sequence generic (non-medical) WikiText corpus, taken from `jlens`'s own `load_wikitext_prompts` — the lens is a function of the weights.
- **Concordance** on the frozen 193-prompt probe set: 50 `general` + 50 `medical` (hand-authored one-hop cloze, style-matched control pairs) + 93 `multihop` from `jlens`'s `lens-eval-multihop.json`.
- **Behavioral** trajectory in parallel: MedQA, MedMCQA, PubMedQA, reported as *answered* accuracy with answer rate alongside. **Scored by an LLM judge**, not a regex — see the instrument slide.

All lens numbers below are the **mean over the last $n/5$ source layers** (the probe's `kl_final` / `top1_final`): 3 layers at 270m, 5 at 1b, 6 at 4b, 9 at 12b. Where the single final layer disagrees, it is called out explicitly — see the read-out caveat. Coverage: 14 / 15 / 14 / 14 lens points (270m / 1b / 4b / 12b) — all four complete. **Behavioral: all four arms are judge-scored, complete, and on one instrument** (14 / 15 / 14 / 14 points incl. t=0), so cross-size behavioral comparison is legitimate for the first time in this deck.

The probe step in detail

Runs in `.venv-jlens` (`transformers` 5.14.1 + `jlens @ 581d398`) — a second env, because the training pin 4.53.2 is hard-incompatible. Imports nothing from `gemma_med`.

Per checkpoint (base prepended as step 0, then `checkpoint-*`):

1. **Load** → `jlens.from_hf(hf, tok)`. Forced text-only `Gemma3ForCausalLM`; any loader leaving a weight missing is *rejected*, not accepted.
2. **Fit** a fresh lens: `jlens.fit(model, corpus, dim_batch=16, max_seq_len=128)` on the frozen 1000-seq WikiText corpus. *Dominates runtime.*
3. **Concordance**: for each of the 193 probes, `lens.apply(model, prompt, positions=[-1])` → per source layer, top-1 agreement and $\mathrm{KL}(\text{model} \,\|\, \text{lens})$ at the last token.
4. **Drift** vs. $t{=}0$, per layer: relative Frobenius $\|J_l - J_l^0\|/\|J_l^0\|$ and cosine.

Invariants that make the trajectory comparable

- **One load path for all points.** Letting `Auto*` pick per-path would read $t{=}0$ through the multimodal wrapper and later points through the text model — two forwards on one trajectory, silently corrupting drift.
- **Lens refit every time.** J is a function of the weights.
- **Corpus and probe set frozen** across all checkpoints and sizes.
- **$t{=}0$ lens cached in fp32**, so a resumed run's drift baseline matches a fresh run's exactly.

Output. One `fsync`'d JSON row per checkpoint in `metrics.jsonl`, plus W&B scalars keyed by training step. Resumable: completed steps are skipped and the fit checkpoints every 50 prompts, so a requeue does not wipe the trajectory.

Reading the metrics — the two drift measures

J_l = lens Jacobian at source layer l; J_l^0 = the same layer in the $t{=}0$ base lens. n = number of source layers (18 / 26 / 34 / 48 at 270m / 1b / 4b / 12b). Both measures are computed per layer, then averaged over all n; both are exactly 0 / 1 at $t{=}0$ by construction.

Metric	How it is computed	Direction
rel. Frobenius `relfro_mean`	$\|J_l - J_l^0\|_F \,/\, \|J_l^0\|_F$ — the *size* of the change, normalised by the base magnitude.	**Neither better nor worse.** It says *how far* the lens moved, not whether the move helped. Larger = more change.
cosine `cos_mean`	$\langle J_l, J_l^0\rangle / (\|J_l\|\,\|J_l^0\|)$, flattened over all entries — the *direction* of the change.	**Neither.** 1 = pointing the same way as the base; lower = more rotation. Bounded above by 1.

Why both. They can move independently: a lens can grow in magnitude without rotating (relfro up, cosine ≈1) or rotate without changing size. In practice they track each other here, which is itself a mild sanity check on the fit.

Not comparable across sizes. J_l has a different shape at every arm, so only these normalised scalars travel — never a raw Jacobian entry.

Reading the metrics — the four faithfulness measures

All four compare the **lens read-out** against the **model's actual next-token distribution**, at the last prompt token, averaged over the prompts of one cue. They differ in which layers they read and how they score.

Metric	How it is computed	Direction
faithfulness KL `kl_final`	$\mathrm{KL}(\text{model} \parallel \text{lens})$, averaged over the **last $n/5$ layers**. *The workhorse metric.*	**Lower is better** — the lens predicts what the model actually says. $0 =$ perfect; unbounded above.
top-1 agreement `top1_final`	Fraction of prompts where $\arg\max$ lens $= \arg\max$ model, same last-$n/5$ layers.	**Higher is better**, range 0–1. Much coarser than KL — see caveats.
`top1_auc`	The same agreement averaged over **all n** layers, not just the top fifth. Despite the name, a mean rather than an integral.	**Higher is better** — agreement holds deeper down the stack, not only near the output.
`crossover_frac`	Shallowest layer whose top-1 agreement reaches 0.5, as a fraction of depth; 1.0 if it never does.	**Lower is better** — the lens locks onto the model's answer earlier in depth.

Cues: `general` / `medical` are the 50+50 hand-authored style-matched pairs; `multihop` the 93 externally sourced prompts. *The cue is the unit of comparison* — medical is read against general, never on its own. This deck reports `kl_final` and `top1_final`; the other two are computed and stored but not used.

Reading the metrics — direction, and what not to do

Behavioral metrics

- **answer rate** — % of generations that committed to an option. Higher better. A pure *format* measure.
- **answered accuracy** — accuracy among those only. Higher better. **This is the knowledge signal.**
- **raw accuracy** — non-answers counted wrong. Higher better, but confounded: it moves when *either* knowledge or formatting moves.

Why it matters: on 270m PubMedQA, raw accuracy *rises* 12.6 → 43.6 — reading as a large gain. Answered accuracy over the same span *falls*, 57.8 → 55.0: all of it is the answer rate going 21.8% → 79.2%. The model learned to *commit to an option*, not to answer correctly. **Raw alone inverts the sign.**

Three rules for reading these numbers

1. **Never compare absolute KL across sizes.** `kl_final` averages a *different number of layers* per arm (3 / 5 / 6 / 9), over models of different depth and width. Only the **within-arm Δ from t=0** is comparable across arms.
2. **Always read against t=0.** Every figure carries the untuned base as a dashed line or a pale bar. A finetuned-only number says nothing.
3. **Name the read-out.** "KL" is ambiguous: the layer mean and the single final layer disagree in sign on 270m medical. This deck is the **layer mean** throughout.

Directional summary. Drift (relfro ↑, cosine ↓) says the lens *changed*; KL ↑ and top-1 ↓ say it *got worse at predicting the model*. The two are independent — a lens can move far and stay faithful, or barely move and lose faithfulness.

Findings

1. **Drift has a sharp onset, then saturates —** and at 4b/12b partly *retreats*. The Jacobian finishes moving in the first $\sim$100 steps. **Onset gets earlier with size**; at 12b it is underway by step 2.

2–3. **Held pending 27b.** Both faithfulness-vs-size findings from the three-arm deck (“the cost scales with size”; “medical is not protected”) **reverse at 12b**. Whether 4b or 12b is the outlier is what 27b decides — so the data are on the slides and the *interpretation* is deliberately not written yet.

4. **Medical SFT works, but the gain is bounded by headroom, not size.** Answered-accuracy Δ on MedQA / MedMCQA / PubMedQA: 270m $-1.9/-10.2/-2.8$, 1b $+4.3/+2.7/+2.8$, 4b $+8.2/+7.0/+13.2$, 12b $+1.0/+6.2/+21.6$. **270m is the only arm that gets worse.**

5. **The timing ordering holds; the correlation behind it does not.** Drift finishes (steps 16–32) long before accuracy moves (2048+) at every size — but drift-vs-answer-rate is only -0.33 to -0.48 on MedQA, and $+0.92$ at 270m on MedMCQA. **The $r \approx -0.9$ in the previous deck was an artifact of `rp`=`1.0`.**

Two things changed at once and must not be conflated: the **12b arm** was added, and the **instrument was re-frozen** (`rp`=`1.1` + a new judge). Finding 5 fell to the instrument; findings 2–3 to 12b.

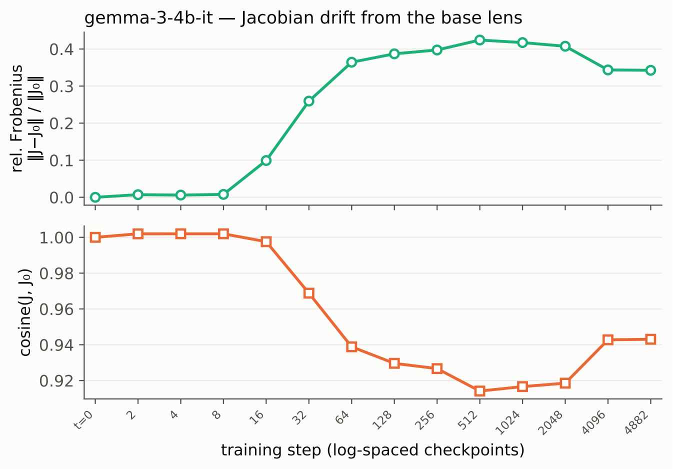
1. Drift onsets sharply, then saturates

	onset (mean > 0.05)	peak relfro	final relfro / cos (layer mean)	per-layer max
`270m-it`	step 64	0.250 @ 1024	0.249 / 0.976	0.350
`1b-it`	step 16	0.308 @ 2048	0.279 / 0.956	0.418
`4b-it`	step 16	0.424 @ 512	0.343 / 0.943	0.531
`12b-it`	step 2	0.472 @ 2048	0.421 / 0.900	0.692

- **Onset is earlier and the swing larger the bigger the model**, now over four sizes: final relfro $0.249 < 0.279 < 0.343 < 0.421$, and cosine falls monotonically $0.976 > 0.956 > 0.943 > 0.900$. This is the one size trend in the deck that 12b *strengthens* rather than breaks.
- The "nothing moves for ∼8 steps" claim is now false. It held at ≤ 0.008 for 270m/1b/4b, but 12b reads 0.055 at step 2 and 0.091 at step 4 — an order of magnitude higher, before a single epoch of anything. Whether that is real early-layer motion or a noisier fit at d_{model}=3840 is **not yet established**.
- **The late partial retreat generalises upward.** 4b peaks 0.424 @ 512 and walks back 19%; 12b peaks 0.472 @ 2048 and walks back 11%. 1b shows a hint (−9%); 270m is flat.

Robust under either the layer mean or the per-layer max. The 12b early-step reading is the one number on this slide to treat as provisional.

1. Drift — `4b-it`



The cosine panel reads slightly *above* 1.0 at steps 2–8. Cosine cannot exceed 1: this is fp32 dot-product accumulation over ∼6.5M entries per layer while the norms use a stabler path. It is visible only where drift is ≈0 and affects no step ≥ 16.

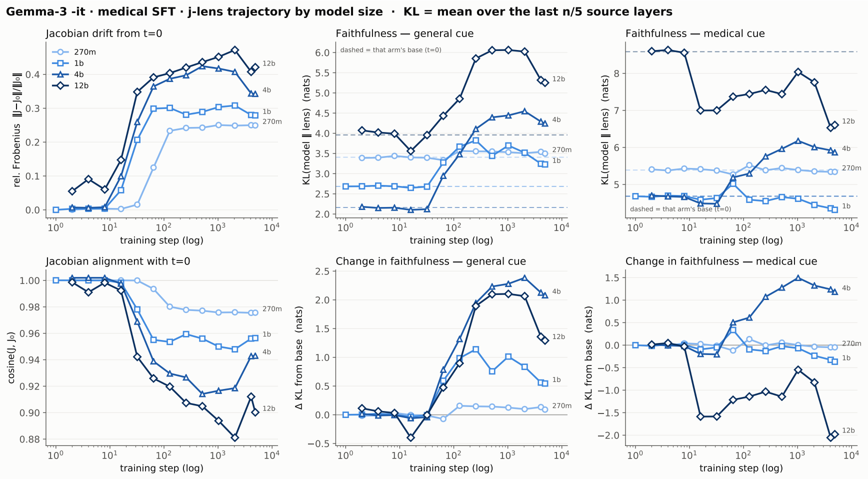
2–3. Faithfulness vs. size — the size trend does not survive 12b

Lens↔model KL, mean over the last $n/5$ layers (the probe's `kl_final`) — *lower = lens predicts the model better*. Base → final checkpoint.

	ΔKL `general`		ΔKL `medical`		ΔKL `multihop` (OOD)	
`270m-it`	3.407 → 3.497	(+0.09)	5.388 → 5.340	(−0.05)	1.958 → 1.975	(+0.02)
`1b-it`	2.684 → 3.229	(+0.54)	4.682 → 4.314	(−0.37)	2.423 → 3.430	(+1.01)
`4b-it`	2.163 → 4.243	(**+2.08**)	4.683 → 5.867	(**+1.19**)	4.088 → 5.715	(+1.63)
`12b-it`	3.959 → 5.251	(+1.29)	8.584 → 6.606	(**−1.98**)	7.267 → 9.578	(**+2.31**)
top-1 `medical`:	270m 0.260 → 0.260		1b 0.448 → 0.572		4b 0.677 → 0.660	12b 0.596 → 0.720

- **"Monotone in size" is dead on `general`**: $+0.09/+0.54/+2.08/+$**1.29** turns over at 12b. Only `multihop` — the one externally-sourced cue — is still monotone.
- **"Medical is not protected" is dead too**, in the other direction: 12b's −1.98 is the largest protective move at any size, and top-1 agrees (+0.124). On four points, 4b is the anomaly.
- **And "bigger models start more faithful" fails at $t=0$**: base `general` KL runs 3.41/2.68/2.16/**3.96**.
- **What is held.** Three readings survive four points: the cost peaks at 4b and falls; 12b is anomalous; or these deltas are not a function of size at all. **27b discriminates the first two** — but a single LR cannot separate "size" from "a fixed lr 1e−5 growing too large," and a non-monotone curve is that confound's signature.

2–3. All four arms on one axis



Arms shaded light→dark by size (an *ordinal* ramp — size is ordered, so it does not take categorical hues). x is the real training step on a log axis, since the arms have different step grids; $t=0$ is each arm's dashed baseline (top row) and the zero line (bottom row), never a point. **The bottom-right panel is the held finding**: 12b (darkest) runs *below* zero for the whole trajectory while 4b runs above it.

2–3. What survives the 12b arm

Both headline size claims reversed, but the trajectory data are not uninformative. One statement holds at **every** size:

	Δ `medical`	Δ `general`	Δ `multihop`	medical is...
`270m-it`	−0.05	+0.09	+0.02	least damaged
`1b-it`	−0.37	+0.54	+1.01	least damaged
`4b-it`	+1.19	+2.08	+1.63	least damaged
`12b-it`	−1.98	+1.29	+2.31	least damaged

- **Medical is the least-damaged cue at all four sizes** — by 0.14 / 0.91 / 0.90 / 3.27 nats over `general`. This is the finding that has held from two arms to four, and it is *directional*, so it does not depend on the cross-size absolute comparison the deck forbids.
- **Medical SFT buys the medical cue relative ground at every size.** Whether it also buys *absolute* ground is what flips: yes at 270m/1b/12b, no at 4b.
- Do not read this as "medical improves." At 4b it degrades; the claim is only that it degrades *less* than everything else.

Read-out check. Layer mean and single final layer agree in sign on all three cues at 4b and 12b. The lone sign flip in this study remains 270m `medical` — see the read-out caveat.

2–3. The domain gap narrows at every size — by three mechanisms

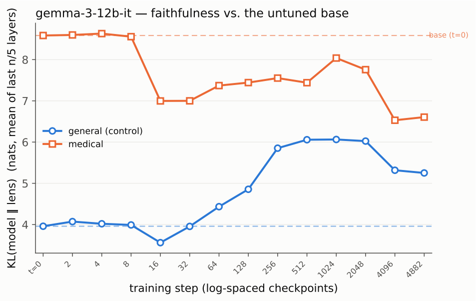
Within each arm the `medical` cue starts 2–4.6 nats less faithful than `general`. **All four arms narrow that gap** — but not by the same mechanism.

	$t=0$	final	Δ	closes by
`270m-it`	1.981	1.843	−0.138	both ends
`1b-it`	1.998	1.085	−0.912	both ends
`4b-it`	2.520	1.624	−0.895	general only
`12b-it`	4.625	1.355	**−3.271**	both ends

At 270m, 1b and 12b the gap closes **from both ends** — medical improves *and* general regresses toward it.

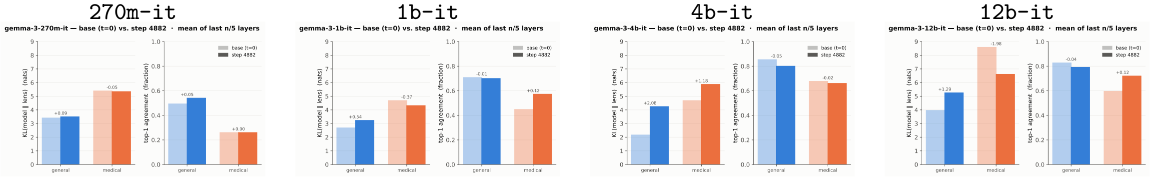
At 4b it closes from one end: both cues degrade, `general` just faster (+2.08 vs. +1.19). **A** −0.9 **gap change means opposite things at 1b and 4b** — which is why the gap is never reported on its own.

12b starts with by far the widest gap (4.63) and closes three quarters of it. `multihop` degrades at *all four* sizes ($+0.02/+1.01/+1.63/+2.31$).



The 12b panel: medical (upper) falls hard toward general (lower), which rises — the “both ends” shape. Contrast the 4b panel, where both curves rise away from their dashed $t=0$ baselines. All four arms are on the previous slide.

2–3. Base vs. finetuned, side by side



Pale bar = untuned base (t=0), solid = step 4882. All four are pinned to the same axes (KL 0–9 nats, top-1 0–1), so the panels are directly superimposable — the range widened from 0–6 to fit 12b's medical base at 8.58. Compare the Δ *labels*, not the bar heights: absolute KL is not comparable across sizes (different depth and width, and a different number of layers in the $n/5$ mean). Read that way, 270m's largest KL move (+0.09) is about a quarter of 1b's smallest (−0.37), and 12b's medical move (−1.98) is the largest at any size in either direction.

4. The instrument, and why every old behavioral number is void

Two things changed at once on 2026-08-05, and both invalidate comparisons to any earlier version of this deck.

1. **Decoding:** `rp=1.0` → `1.1`
   - The old answer-rate collapse was **verbatim repetition loops under greedy decoding**, not chain-of-thought overrunning the token cap. At 4b step-4882, 311 of 327 unanswered MedMCQA rows were loops; 8 were real truncation.
   - A repetition penalty fixes it outright: 4b MedQA answer rate now never leaves 97.9–99.8% across the whole trajectory (it bottomed at 83.8% before).
   - No capability appears. `acc_answered` is essentially unchanged — the knowledge was never missing, only the format.

2. **Judge: Claude Haiku → self-hosted Kimi**
   - Scoring is an **LLM judge** (`gemma_med.judge`), not a regex: it matches option *content*, not the letter, and is told the gold answer and forbidden to re-derive it.
   - Judge *decode* is now frozen too — `temperature=0.0, seed=0`, recorded as `judge_decode`. Before this date each provider sampled at its own default, so "the judge" was frozen at the model level only.
   - All four arms are now judge-scored by the same judge, which is what makes the cross-size table on the next slide legitimate.

Both changes are cross-validated. Judge: on 300 identical MedQA rows at 4b step-1024, Claude Haiku and Kimi agree to within 1.3 pp raw (45.0 vs. 46.3). Judge decode: re-judging the same rows twice flipped `chosen` on 1.34% under the old wire vs. 0.33% at $t=0$, so the old config moved `acc_raw` by up to 1.0 pp *between two identical runs*; old-vs-new differs by ≤ 0.67 pp, i.e. inside that noise. Freezing buys reproducibility, not a different answer. $t=0$ is still not bitwise deterministic — batch composition moves floating-point ties.

4. Behavior — all four arms, one instrument

Judge-scored at `rp=1.1`. **Answered** accuracy, with answer rate in parentheses. t=0 is each arm's untuned base.

	MedQA		MedMCQA		PubMedQA	
`270m-it` t=0	26.7	(96.5%)	36.9	(59.3%)	57.8	(21.8%)
`270m-it` final	24.8	(89.5%)	26.6	(92.5%)	55.0	(79.2%)
Δ	−1.9		−10.2		−2.8	
`1b-it` t=0	31.1	(99.9%)	34.3	(99.9%)	51.6	(100%)
`1b-it` final	35.3	(99.0%)	37.0	(99.0%)	54.4	(99.6%)
Δ	+4.3		+2.7		+2.8	
`4b-it` t=0	50.9	(99.9%)	46.6	(99.8%)	60.0	(100%)
`4b-it` final	**59.2**	(99.5%)	**53.6**	(99.5%)	**73.2**	(100%)
Δ	+8.2		+7.0		+13.2	
`12b-it` t=0	70.0	(100%)	58.8	(100%)	56.4	(100%)
`12b-it` final	**71.0**	(98.8%)	**65.0**	(99.5%)	**78.0**	(100%)
Δ	+1.0		+6.2		+21.6	

- **270m is the only arm that gets worse** — and not by format churn: its answer rate *improves* on two of three benchmarks while answered accuracy falls on all three.
- **The gain is bounded by headroom, not size.** 12b gains +1.0 on MedQA from a base of 70.0, but +21.6 on PubMedQA from 56.4 — the largest single gain in the study.

4. A second failure mode: collapse onto the majority label

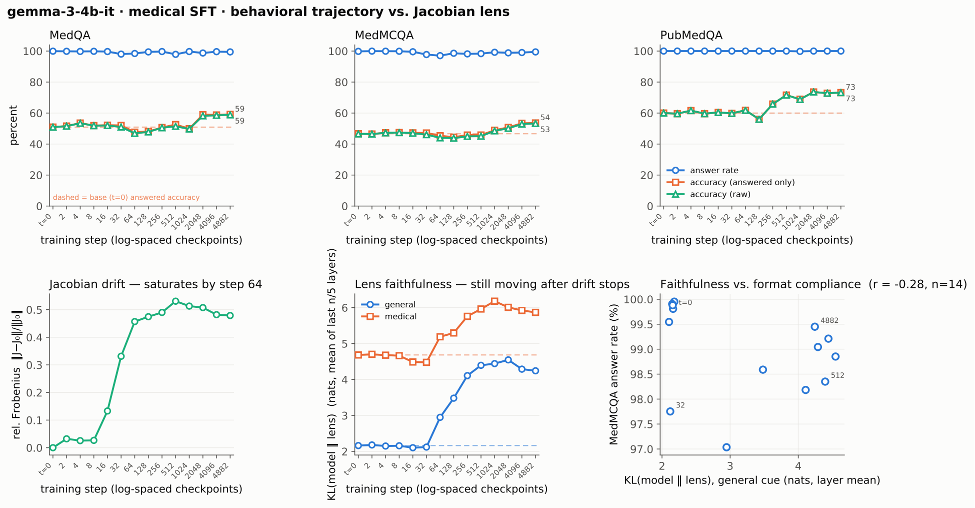
Format collapse is not the only thing that goes wrong mid-training. Share of *answered* items assigned to the dominant label, against that label's prevalence in the gold answers:

	MedQA ("A")		MedMCQA ("A")		PubMedQA ("yes")	
	chosen	gold	chosen	gold	chosen	gold
t=0	18.5	27.7	21.6	32.2	50.4	55.2
step 512	28.6	27.7	37.5	32.2	65.2	55.2
step 1024	**37.7**	27.7	**39.3**	32.2	69.7	55.2
final	29.5	27.7	33.1	32.2	66.8	55.2

- **The base model is *anti*-majority** (18.5 vs. 27.7). SFT pushes it through neutral and overshoots.
- **Step 1024 is a spike, not a trend** — it recovers. Worth knowing before building a story on any single checkpoint.
- **PubMedQA is the control that reframes it.** It has no "A" to collapse onto, yet shows the same overshoot on "yes". So this is **majority-label over-commitment, not position bias.**

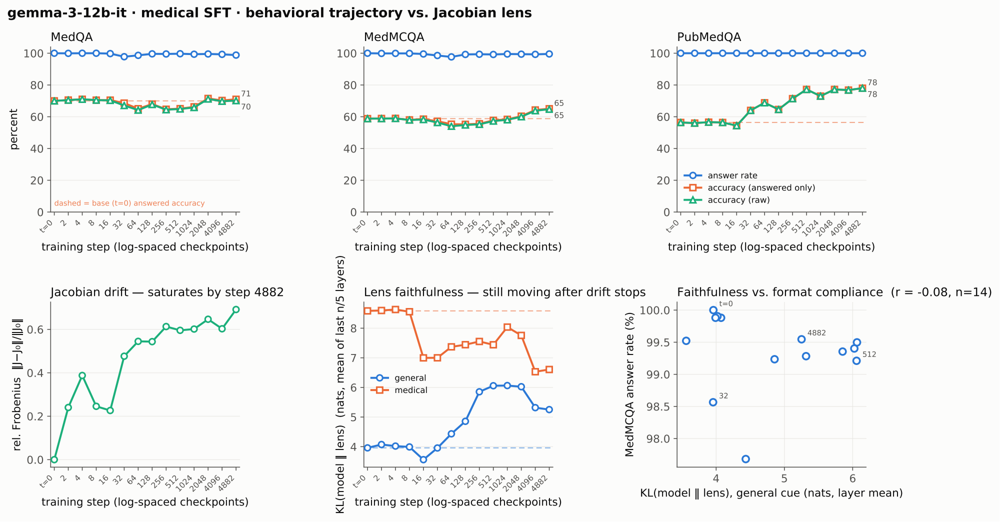
This mode survives the instrument change — recomputed at `rp=1.1`, it is the same shape, roughly two-thirds the amplitude (the MedQA step-1024 spike was 57.2 at `rp=1.0`, 37.7 now). Unlike the format-collapse mode, it is not a decoding artifact. Note this means the two failure modes no longer trade off: with format compliance held near 100% throughout, over-commitment is now the *only* mid-training pathology visible at 4b.

4. Joined behavioral + lens trajectory — `4b-it`

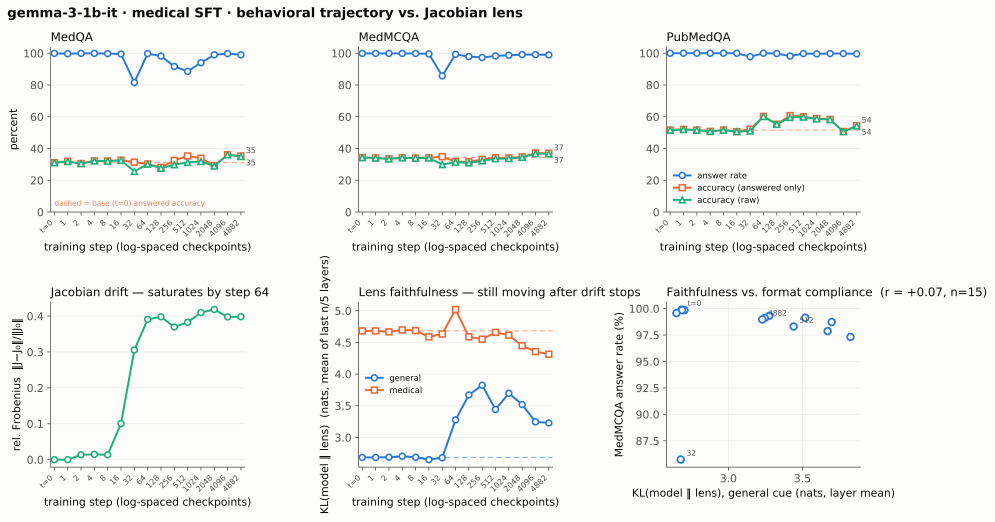


Judge-scored at `rp=1.1`. Compare this against the previous version of the deck: answer rate (blue) now sits flat near the top instead of diverging from answered accuracy (orange) at step 32. The format/knowledge split that this panel used to make visible was the decoding.

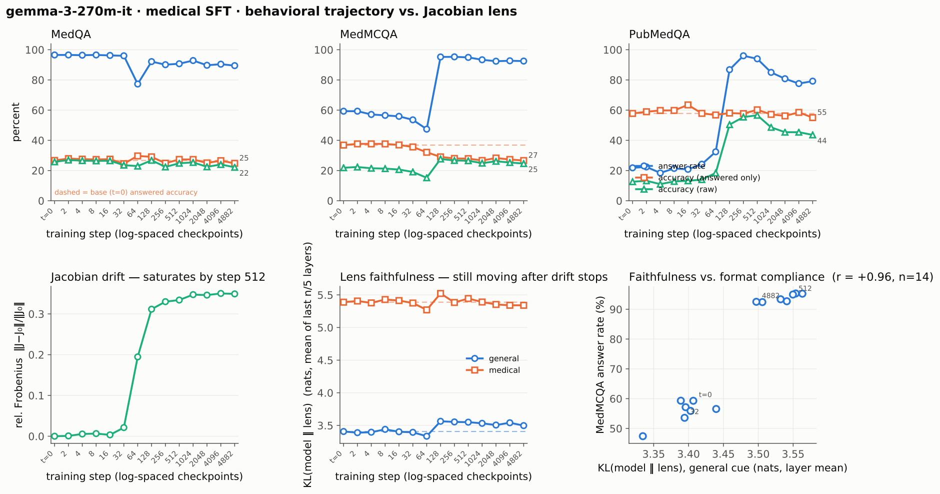
4. Joined behavioral + lens trajectory — `12b-it`



4. Joined behavioral + lens trajectory — `1b-it`



4. Joined behavioral + lens trajectory — `270m-it`



The KL panel is the layer mean, as everywhere else in this deck; at the single final layer the medical curve rises instead. 270m is the one arm where answer rate still moves a lot under `rp=1.1` — it *rises* (MedMCQA 59 → 92%), which is why its drift↔rate correlation comes out *positive*.

5. Timing holds; the correlation behind it does not

The ordering is the robust part, and it holds at all four sizes — `4b-it` shown:

Signal	Onset	Notes
Jacobian drift	steps 16–32	0.099 → 0.259 (layer mean); plateaus by step 64
Task accuracy	steps 2048+	MedQA answered 52.6 at step 512; 59 only at 2048–4882

But "drift leads format collapse" does not survive `rp=1.1` — Pearson r vs. drift (layer mean), answer rate / answered accuracy:

	MedQA	MedMCQA	PubMedQA
`270m-it`	−0.49 / −0.22	**+0.92** / −0.99	+0.97 / −0.56
`1b-it`	−0.33 / +0.11	−0.20 / +0.05	−0.31 / +0.73
`4b-it`	−0.47 / +0.06	−0.65 / +0.15	−0.27 / +0.57
`12b-it`	−0.48 / −0.49	−0.48 / +0.07	— / +0.90

- The old −**0.91** on 4b MedMCQA is now −0.65, and the sign flips outright at 270m (+0.92) — there answer rate *rises* over training. (PubMedQA at 12b is `--`: answer rate is exactly 100.0% at every checkpoint, so r is undefined.)
- **With format compliance pinned near** 100% **by the decoding, there is almost no variance in answer rate left to correlate with.** The old r was measuring an artifact of greedy decoding.
- **What still stands:** the Jacobian finishes moving before the model learns any medicine. What does not: that it moves *with* format collapse specifically.

Caveats — the measurements

- **Top-1 is coarse.** n=50 prompts for `general/medical` (93 for `multihop`), so *per-layer* top-1 quantizes at 0.02; the reported statistic averages 3 / 5 / 6 / 9 layers, giving a 1/150–1/450 grain that makes small moves look more precise than they are. Single-checkpoint wiggles of $\pm$0.04 are still noise. *Trust the KL trends, not top-1.*
- **Read-out choice is not innocent.** Layer-mean and final-layer KL disagree in sign on 270m medical. Every claim here uses the layer mean; an earlier draft of this deck used the final layer and reported a 270m$\leftrightarrow$1b *inversion* that does not survive. Report the statistic, not just the number. (It does not recur at 4b.)
- **A numerical artifact in the drift cosine.** At 4b steps 2–8, `cos_mean` reads 1.001–1.003; cosine cannot exceed 1. The fp32 `dot` over $\sim$6.5M entries per layer is biased against the stabler norm computation. Only visible where drift $\approx$0; no reported claim depends on it, but `jac_drift` should accumulate in fp64.
- **The `general/medical` cues are hand-authored**, not sampled from a corpus — the domain-gap result inherits whatever bias is in those 100 prompts. `multihop` is the one cue with an external, independently-constructed source.

Caveats — the scope

- **Four sizes is still not a scaling curve.** The 12b arm did not extend the trend it was run to test — it reversed it, on both `general` and `medical`. Four points spanning $\sim 44\times$ in parameters now show one non-monotone cue, one monotone cue, and a medical cue whose sign alternates. **27b arbitrates 4b-vs-12b but cannot make four noisy points into a curve.**

- **One hyperparameter setting — now the load-bearing caveat.** All four arms are full FT at lr $1\mathrm{e}{-5}$ on one mixture. Whether these deltas are about *size*, or about a fixed LR being progressively too large as the model grows, is not separable from these runs — and **a non-monotone `general` curve is exactly that confound's signature**. The decisive experiment is a second LR at one size, not a fifth size.

- **The behavioral instrument is finally uniform** — four arms, one decode config, one judge, judged $t{=}0$ baselines for all. Cross-size behavioral comparison is legitimate for the first time. The cost: every behavioral number predating 2026-08-05 is void, and finding 5 did not survive the change.

- **Coverage gaps.** 4b, 270m and 12b have no `checkpoint-1`, so their grids start at step 2; only 1b probed step 1. No `-pt` arm exists at any size — the pt-vs-it starting-point axis is entirely unstarted.

- **The 12b early-step drift is unexplained.** 0.055 at step 2 against ≤ 0.008 for the three smaller arms. Real early motion or a noisier fit at $d_{\text{model}}{=}3840$ — not yet distinguished, and it is the one number in finding 1 that is provisional.

Next

1. **Finish 27b** and settle findings 2–3. The behavioral half is already decoded and judged at both `rp` values; the lens is 8/13 checkpoints in, at $\sim$34 h per checkpoint. **This is the only blocker on un-holding the two findings.**
2. **A second LR at one size** — 4b or 12b at, say, $3\mathrm{e}{-6}$. Now the highest-value experiment in the study: it is the only thing that separates "the cost is about size" from "the cost is about a fixed LR being too large for a bigger model," and the 12b reversal is what makes that ambiguity load-bearing rather than pedantic.
3. **Widen the `general`/`medical` cues** beyond the 100 hand-authored prompts, to check the domain-gap collapse is not an artifact of how those pairs were written. 12b's 4.63-nat starting gap makes this more pressing, not less.
4. **Open a `-pt` arm** at 1b: does a non-aligned starting point show the same drift-before-competence ordering?
5. **Anchor against `medgemma-27b-text-it`** on the same probe set to place the trajectory endpoints on an absolute scale.

Answered since the last version: **270m and 1b are re-judged**, so all four arms share one instrument; **the 12b arm exists** end to end. Retracted since the last version: the size-monotone faithfulness cost (finding 2), medical-is-not-protected (finding 3), and drift-leads-format-collapse at $r = -0.91$ (finding 5) — the first two by 12b, the third by the decode fix.

Data provenance

Artifact	Path
Lens metrics	eval/jlens/{270m_it_full, 1b_it_full_v3, 4b_it_full, 12b_it_full}/metrics.jsonl (14/15/14/14 rows)
Joined traj.	eval/traj/<arm>_rp11/joined.json (all judge-scored)
t=0 dirs	eval/gemma-3-{270m,1b,4b,12b}-it-baseline-rp11/
Fit corpus	data/jlens/fit_corpus_v2.txt (1000 WikiText seq, frozen)
Probe set	data/jlens/probe_set_v2.tsv (50/50/93, frozen)
Figures	figs/<lens-tag>/ lens (decode-independent); figs/<arm>_rp11/fig_joined.pdf (decode-dependent)
Probe jobs	25776556 (1b), 25799677 (270m), 25948088–25948100 (4b), 13-way fanout (12b)

All four runs logged fit_corpus=1000 probes=193 — the frozen v2 inputs. The v1 files (40 seq / 30 prompts) are legacy, still in data/jlens/, used only by the superseded runs.

4b and 12b were probed one Slurm job per checkpoint (4b ∼3.5 h each; 12b longer — fit cost scales as $d_{\text{model}} \times$ params), all 13 workers reusing the *same* fp32 t=0 lens so drift shares one baseline; rows merged by scripts/merge_jlens_fanout.py, which refuses the merge if any non-base step appears twice.

Superseded: 4b_it_validate{,2} (wrapper-validation spikes), 1b_it_full_v2 (v1 30-prompt set), 1b_it_full_probev2, 270m_verify (Phase-0 smoke test).

Data provenance — the behavioral instrument (all four arms)

Field	Value
Decode	`temperature=0.0, max_new_tokens=1024,` `repetition_penalty=1.1, seed=0` **frozen 2026-08-05**
Scorer	`gemma_med.judge` (replaces `answer_parsing.py`)
Provider	`local` — the lab's self-hosted Kimi, BigPurple only
Judge model	`Barney` **frozen for all four trajectories**
Judge decode	`temperature=0.0, seed=0` **also frozen 2026-08-05**
Verdicts	`correct` / `incorrect` / `no_answer`, schema-constrained
Judgments	`eval/traj/*_rp11/step-*/*_judgments.jsonl`
Provenance	`.../step-*/judged_summary.json` (13/14/13/13 files)

A second decode is on disk and must not be mixed in. Every arm was also decoded at `rp=1.05` (`eval/traj/*_rp105/`), kept as a sensitivity check: it halves the repetition loops while costing less `acc_answered` than 1.10 does, and it preserves a muted version of the old answer-rate dip (4b MedQA 92.9% at step 512 vs. 97.9% at `rp=1.1`). Different instrument — never in the same figure.

`judge_self_disagreement` — **a quality signal that scores nothing.** It counts rows where the judge's verdict string conflicts with the option it picked: 0.05–0.25% on these runs. It changes no number — `correct` and `answered` are both recomputed from `chosen`, never from `verdict`. It ran higher (0.95% overall, peaking 2.8%) on the `rp=1.0` corpus, where the judged model rambled more; the statistic tracks **how rambly the judged model is**, not how confused the judge is.